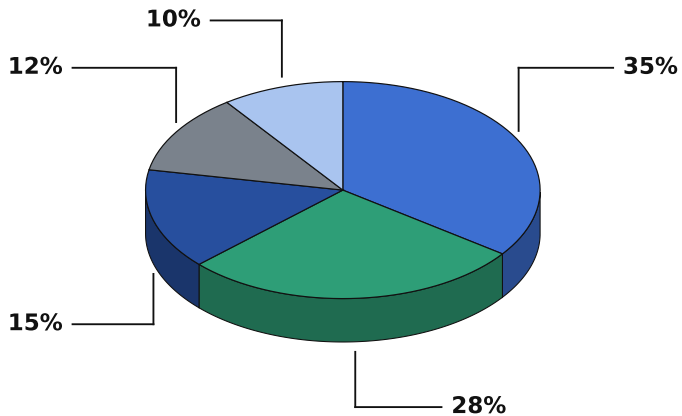




one decode step $\approx 1,080$ ms (instrumented, 16k)

low-rank SVD scoring
35% · ≈ 378 ms

residual attention
28% · ≈ 302 ms

cache merge
15% · ≈ 162 ms

SRL query routing
12% · ≈ 130 ms

fused-buffer materialisation
10% · ≈ 108 ms

63% of the step is the per-token low-rank contraction $q \cdot V_K$ plus the residual attend — not buffer materialisation. Collapsing the two into one fused launch is the open optimisation.